

Trainings ▾

General Information

with Mechanically Actuated Injector

Only plain cam follower covers or covers with a pipe tapped hole for step timing control (STC) oil drain-back are on K38 and K50 engines. Those covers with provisions for oil fill or crankcase breather are **only** on the K19.

A revised cover gasket was released in February, 1984, engine serial number (ESN) 33108281.

All cam follower covers are manufactured from aluminum. There are two types of cam follower gaskets. The earlier gaskets have a bead of silicone sealant that **must** be installed with the bead touching the cover. The present gaskets are manufactured from a material that enlarges after contacting the oil. The use of shellac or cement on the new gaskets reduces the sealing quality of the gaskets.

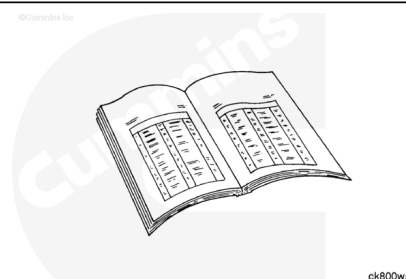
The K2000E, K1800E, and K1500E engines use a cam follower cover without internal ribs to provide clearance for the flange head cam follower mounting capscrews.

Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

When the engine is operating with the rocker lever covers removed, oil splashing can cause personal injury. To reduce the possibility of personal injury, wear eye and face protection.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the lubricating oil from the engine. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Remove the crankcase breather. Refer to Procedure 003-001 in Section 3 (</qs3/pubsys2/xml/en/procedures/28/28-003-001-tr.html>).
- Remove the left and right bank wiring harness. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/05/05-019-043.html>) in the Troubleshooting and Repair Manual Electronic Control System, QSK50 and QSK60 Modular Common Rail System, Bulletin 4021533.
- Remove the left and right bank injector and exhaust gas temperature sensor harness. Refer to Procedure 019-043 in Section 19 (</qs3/pubsys2/xml/en/procedures/05/05-019-043.html>).

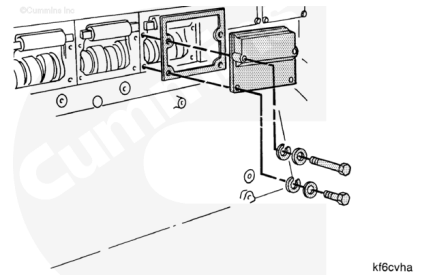
Remove

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Note : If the engine contains an STC system, two of the cam follower covers (one on each bank) will contain a flexible hose fitting. Make a note of the location of these covers.

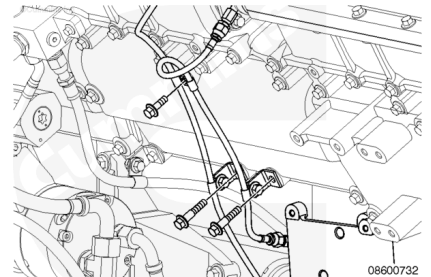
Remove the four capscrews, covers, and gaskets.

Discard the gaskets.



with Electronically Actuated Injector

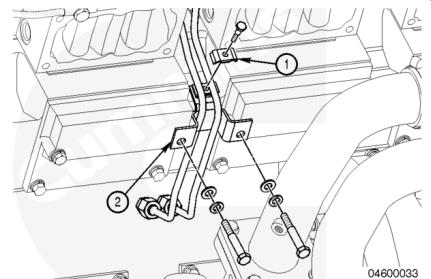
Remove both the capscrews and P-clips securing the electronic control module (ECM) cooling plate lines to the engine on the left bank.



Remove the capscrew and upper part of the bracket (1).

Remove the capscrews, locking washers, and plain washers securing the bracket (2) on the second and third cam follower cover on the left and right banks.

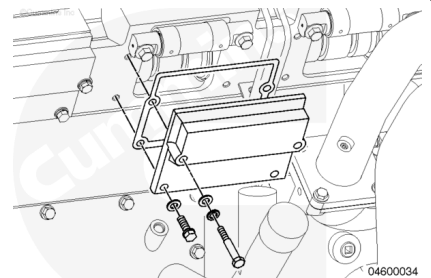
Slide the bracket down to clear the cam follower covers.



Note : The upper capscrews (long) have loose locking and plain washers. The lower capscrews (short) have captive locking washers and loose plain washers.

Remove the capscrews, locking washers, and plain washers securing the cam follower covers to the engine.

Remove the cam follower cover and gasket. Save the gasket for possible reuse.

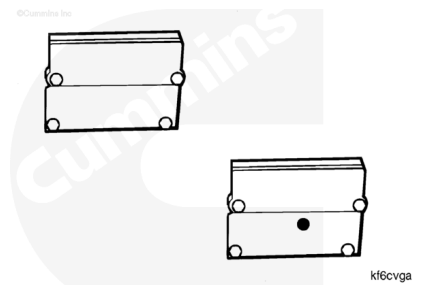


Clean and Inspect for Reuse

with Mechanically Actuated Injector

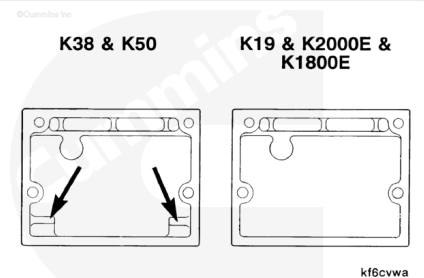
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use a solvent that will **not** harm aluminum. Clean the covers.
Check for covers for cracks or damage to the gasket sealing face.

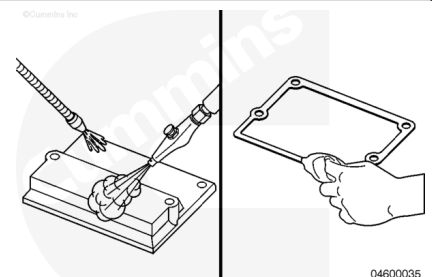
Check covers for cracks or damage to the gasket sealing face.
Replace cracked or damaged covers.



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⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the cam follower covers.

Dry with compressed air.

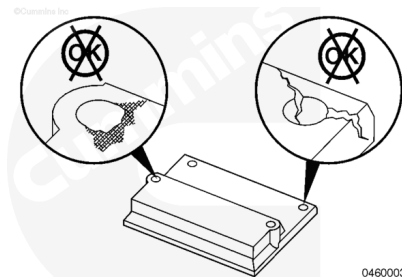
Do **not** use steam or cleaning solvents on the gasket. Use a clean cloth and wipe the gasket clean.

Inspect the cam follower covers for nicks, burrs, cracks, or other damage.

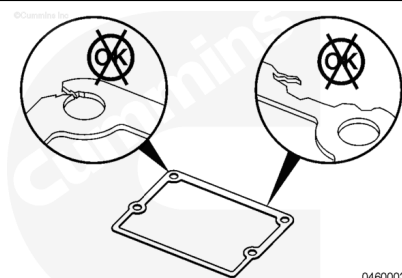
Use a fine crocus cloth to remove small nicks and burrs.

Inspect the threaded capscrew holes for damage.

Replace the cam follower cover if there is any damage that can cause a leak.



Inspect the cam follower cover gasket for any signs of damage, tears, and cracks. Replace the gasket if any sign of damage is found.



Install

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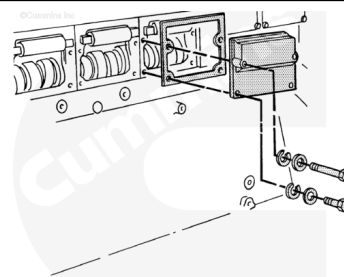
Note : Two different types of gaskets have been used for the cam followers. The older gaskets contain a raised bead of sealant. The older gaskets **must** be installed so that the bead of sealant touches the cover. The revised gaskets do not have a bead and are manufactured from a material that swells when exposed to engine oil. Do not use a gasket adhesive on the revised gasket.

Note : Engines equipped with STC require a cover with a hole for a fitting to be installed on both engine banks. On a K38 engine, install the covers over the followers for the number 4 cylinder. On a K50 engine, install the covers over the followers for the number 6 cylinders.

Install the parts. Use new gaskets.

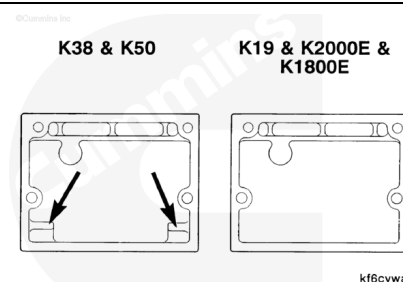
- Gasket
- Cover
- Lock washers (four)
- Flat washers (four)
- Capscrews (four)

Torque Value: 40 n•m [30 ft-lb]



A revised cam follower cover is required for K2000E and K1800E engines. This cover provides a clearance between the flange head cam follower shaft mounting capscrew and the cam covers.

This cover was previously used on K19 engines and does **not** have an internal rib like the K38 and K50 cover.

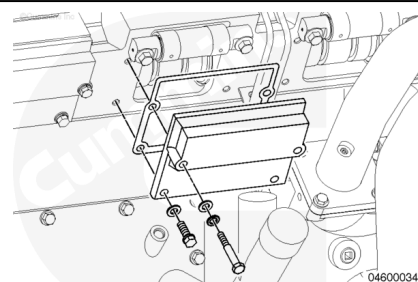


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Install the gasket onto the cam follower cover and install the gasket, cover, and capscrews.

Note : The upper capscrews (1) have loose locking and plain washers and the lower capscrews (2) have captive locking washers and loose plain washers.

Torque Value: 40 n•m [30 ft-lb]



⚠ CAUTION ⚠

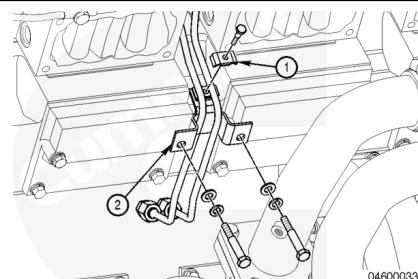
Do not over torque the bracket capscrews. Damage to the bracket can result.

Slide the fuel drain manifold bracket (2) upward into place and install both capscrews, locking washers, and plain washers to the second and third cam follower covers.

Torque Value: 40 n•m [30 ft-lb]

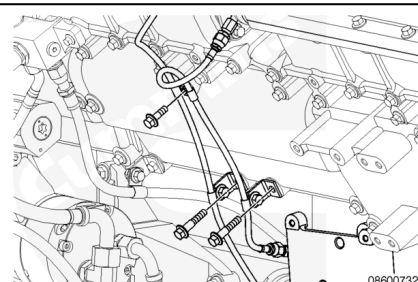
Install the upper part of the bracket (1) that holds the main fuel drain line manifold.

Torque Value: 5 n•m [44 in-lb]



Install the P-clips and capscrews that secure to the ECM cooling plate lines on the left bank of the engine.

Torque Value: 45 n•m [33 ft-lb]

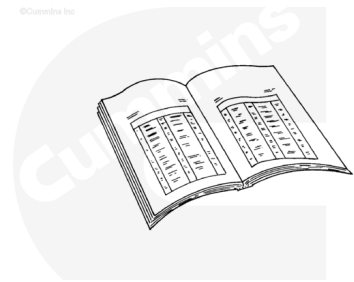


Finishing Steps

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⚠ WARNING ⚠

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- Install the left and right bank injector and EGTS harness. Refer to Procedure 019-043 in Section 19 (</qs3/pubsys2/xml/en/procedures/05/05-019-043.html>) in the Troubleshooting and Repair Manual Electronic Control System, QSK50 and QSK60 Modular Common Rail System, Bulletin 4021533.
- Install the left and right bank wiring harnesses. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/05/05-019-043.html>) in the Troubleshooting and Repair Manual Electronic Control System, QSK50 and QSK60 Modular Common Rail System, Bulletin 4021533.
- Install the crankcase breather. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/28/28-003-001-tr.html>).
- Fill the lubricating oil from the engine. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr-gc.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for leaks.